

COM661 — Back-End API Development

API Endpoints Summary

Base URL	http://127.0.0.1:5001
Authentication	Basic Auth (login) + JWT Bearer Token (all other routes)
Database	MongoDB — carbonDB
Total Endpoints	18 endpoints across 3 route files

1. Authentication & User Management

Method	Endpoint	Description	Auth
POST	/users/register	Register a new user	None
POST	/users/login	Login — returns JWT token	Basic Auth
POST	/users/refresh	Refresh JWT access token	Refresh Token
GET	/users	Get all users (admin only)	JWT Admin
GET	/users/	Get one user	JWT
PUT	/users/	Update user details	JWT
DELETE	/users/	Delete a user	JWT

2. Activity CRUD

Method	Endpoint	Description	Auth
POST	/activities	Create a new activity	JWT
GET	/activities	Get all activities (RBAC)	JWT
GET	/activities/	Get one activity	JWT
PUT	/activities/	Update an activity	JWT
DELETE	/activities/	Delete an activity	JWT

3. Sources Sub-Documents

Method	Endpoint	Description	Auth
POST	/activities//sources	Add a source to activity	JWT

PUT	/activities/sources/	Update a source	JWT
DELETE	/activities/sources/	Delete a source	JWT

4. Filtering & Sorting

Method	Endpoint	Description	Auth
GET	/activities/filter/type?activity_type=...	Filter by activity type	JWT
GET	/activities/filter/date?start_date=...&end_date=...	Filter by date range	JWT
GET	/activities/filter/emission?min_emission=...	Filter by min emission	JWT
GET	/activities/top-emitting	Get top emitting activities	JWT

5. Analytics (MongoDB Aggregation)

Method	Endpoint	Description	Auth
GET	/analytics/total-emissions	Total emissions per user	JWT
GET	/analytics/emissions-by-type	Emissions grouped by type	JWT
GET	/analytics/monthly-trends	Monthly emission trends	JWT
GET	/analytics/highest-activities	Highest emitting activity types	JWT

6. Role-Based Access Control (RBAC)

Role	Activities	Users
admin	Can see and manage ALL activities	Can see and manage ALL users
user	Can only see and manage OWN activities	Can only manage OWN account